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|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Answer: C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|   | <p data-bbox="471 156 875 246">+ ↓: <math>s_y = \cancel{u_y t} + \frac{1}{2}gt^2</math></p> <p data-bbox="471 270 716 319">+ →: <math>s_x = u_x t</math></p> <p data-bbox="471 342 1644 474">Combining the 2 equations gives <math>s_y = \frac{1}{2}g\left(\frac{s_x}{u_x}\right)^2</math> or <math>s_x \propto u_x \sqrt{s_y}</math></p> <p data-bbox="471 487 566 526">Thus,</p> $\frac{x_A}{x_B} = \frac{u_A}{u_B} \sqrt{\frac{s_A}{s_B}}$ $= \frac{2}{1} \sqrt{\frac{1}{2}} = \sqrt{2} = 1.41$ |

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| 4 | Answer: D |
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